

A coupled finite element scheme to study the grinding force and warpage of silicon wafers during the backside grinding process

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ABSTRACT

In this study, a finite element model is developed to analyse the grinding force and warpage of silicon wafers during the backside grinding process. Due to the decreasing size of consumer electronic devices, such as smartphones, notebooks and portable electronics, it is necessary to address the issues pertaining to grinding silicon wafers. The backside grinding process is a mature technology that is widely used for silicon wafers. However, for ultrathin silicon wafers, warpage is a critical issue. Wafer warpage is induced by the residual stress and surface damage that arises during the backside grinding process. To analyse the grinding stress on silicon wafers during the backside grinding process, a finite element model is established by setting dynamic loads and contact conditions. An explicit dynamic model is used to simulate the relationship between the grinding wheel and the silicon wafer. A static model is incorporated with the explicit dynamic model to predict wafer warpage. The simulation results for the residual stress are in good agreement with the experimental results. The results indicate that the wheel rotational speed, wafer rotational speed and feed rate effectively control wafer warpage. Hence, the warpage of ultrathin silicon wafers can be decreased by adjusting the manufacturing process parameters. Furthermore, the developed simulation model can also be used to analyse warpage in fan-out wafers during the backside grinding process.

KEYWORDS: warpage, self-rotating grinding, grinding force, silicon wafer

1. INTRODUCTION

The backside grinding process is widely used for silicon wafers and fan-out wafers to reduce the thickness of microelectronic packages and microelectromechanical systems. It is also an effective and efficient way to decrease the weight of the microelectronic package. However, thin silicon wafers and fan-out wafers have warpage issues due to residual stress and surface damage. The stiffness of silicon wafers and fan-out wafers cannot support the grinding stress on the grinding surface, which results in warpage. Therefore, it is important to determine the residual stress in the silicon wafer during the backside grinding process (Fig. 1). Sun *et al.* [1] used the Rayleigh probability density function to characterize the depth of grinding grooves. They built a theoretical model to determine the relationship between the roughness and the groove depth. Their experimental results matched the theoretical model and were verified by discussing the effects of wheel rotational speed, wafer rotational speed, feed rate and wafer radius. Wafer rotation speed is determined as the number of wafer rotations around the axis of the wafer per minute; wheel rotation speed is the number of wheel rotations around the axis of the wafer per minute and feed rate is the velocity at which the wheel cutter is fed. Zhou *et al.* [2] analysed the grinding process of a thin silicon wafer by using an ultra-precision grinding machine. They found that the

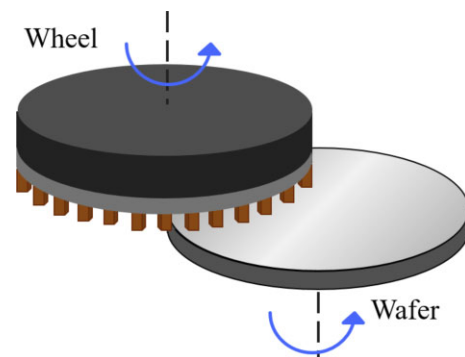


Figure 1 The structure of a backside grinding machine.

abrasive grain depth was an important factor affecting the grinding surface integrity and the surface damage. Paehler *et al.* [3] used a laser-acoustic method based on surface acoustic waves to determine changes in the Young's modulus of monocrystalline silicon wafers caused by subsurface damage. Haapalinna *et al.* [4] studied several measurement methods to evaluate the subsurface damage in silicon wafers. They found that a combined method could determine the damage type in the grinding process and the ductile cutting mode in the silicon material. Pei and Strassbaugh [5] presented an experimental design to investigate the

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fine grinding of silicon wafers. The wheel rotational speed, chuck rotational speed and feed rate were investigated to determine the interaction effects by using a three-factor two-level full factorial design. Gao *et al.* [6] investigated edge chipping sustained by silicon wafers during the grinding process. They utilized machining mechanics and energy theory to demonstrate the mechanisms of edge chipping. Their experimental results revealed that the abrasive grain size, feed rate and wafer thickness were the primary factors responsible for increasing edge chipping.

Raman spectroscopy measurements, theoretical models and finite element models have been used to analyse the grinding force on silicon wafers. Chen and Wolf [7] directly measured the stress on a silicon wafer by using Raman spectroscopy. Additionally, they developed an analytical model to calculate the stress propagation with respect to the depth from the surface. They found that Raman spectroscopy could accurately measure the stress on a silicon wafer up to a depth of 20 μm from the surface. Sun *et al.* [8] proposed a theoretical model to predict the grinding force in silicon wafers to control their thinning quality. The sliding and cutting forces were obtained considering the normal and tangential directions. Their results showed that the grinding force at the edge of the wafer was greater than that at the center of the wafer. Young *et al.* [9] derived a fundamental theory related to the abrasive grain depth-of-cut and various grinding parameters. In their experiment, the critical cut depth was within a range of 20–40 nm. The abrasive grain depth-of-cut varied under different grinding conditions. Gao *et al.* [10] established a mathematical model to analyse the mechanical stress on the damaged layer of silicon wafers based on elasticity theory. The vacuum resultant force, which includes both the supporting force and the attracting force, was considered in the model. They conducted experimental tests to validate the theoretical model, and their experimental results showed that the wafer sustained substantial warpage when the wafer thickness was reduced by less than 200 μm . Tian *et al.* [11] developed a two-dimensional axisymmetric finite element model to predict warpage in ultrathin wafer grinding. The residual stress in the silicon wafer was simulated by applying the strain on the damaged layer. The results from their finite element model were in good agreement with the experimental results. Several studies have used Raman spectrometry, theoretical models and finite element models to calculate the residual stress in the damaged layer of silicon wafers and predict wafer warpage.

In this study, an explicit dynamic model of grinding simulation is established to predict the stress on the wafer surface. The stress predicted by the explicit dynamic model is applied to the damage layer of the static wafer model. Therefore, the explicit dynamic model and the static wafer model are used to predict the warpage of silicon wafers and can be applied to fan-out wafers.

2. BACKSIDE GRINDING PROCESS

During the backside grinding process, the grinding wheel and silicon wafer rotate at the same time, and downward displacement is applied on the grinding wheel through the gear on the grinding wheel to decrease the thickness of the wafer. The wafer grinding process time is dependent on many process parameters and can vary from tens of seconds to several minutes. A schematic dia-

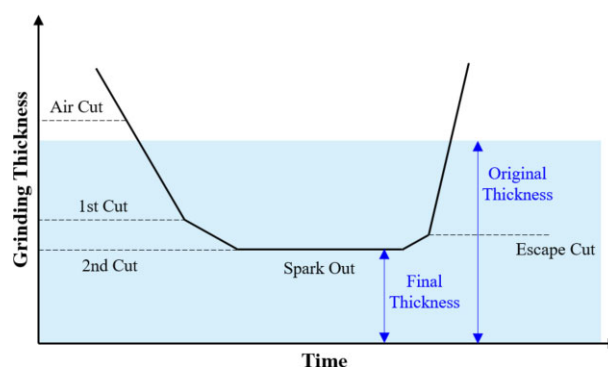


Figure 2 Schematic diagram of the backside grinding process.

gram of the grinding process is presented in Fig. 2. The backside grinding process includes the following steps:

1. Air cut step: Before the grinding wheel touches the wafer, it rotates at a certain initial speed to prevent the grinding wheel from generating excessive grinding stress on the wafer. After the grinding wheel comes into contact with the wafer, it will gradually press down on the wafer until the thickness of the wafer decreases to the specified thickness.
2. First cut: The first cut is the rough grinding process. To avoid excessive damage to the wafer surface during the grinding process, grinding is divided into several stages.
3. Second cut: The second cut is the fine grinding process. In this process, a grinding wheel with a smaller abrasive grain size is used to reduce the damage on the wafer surface, thereby creating a wafer with better smoothness.
4. Escape cut: The grinding wheel gradually moves away from the wafer.

3. METHODOLOGY

This study is divided into the development of two models: the explicit dynamic model and the static state model, as shown in Fig. 3. The explicit dynamic model is used to simulate the actual backside grinding process for calculating the grinding stress on the damaged layer of the silicon wafer. In this study, the explicit dynamic simulation model will be initially established. This dynamic model is created based on the actual manufacturing process and is aimed to predict the values of residual stress on the surface of the wafer at various time points during the grinding process. Subsequently, using the residual stress prediction model proposed by the Stoney equation, the residual stress loading is set to predict the wafer warpage for a specific thickness. Based on this theory, the present study establishes a steady-state simulation model. This model utilizes the predictions of residual stress values obtained from the dynamic simulation model and converts them into predictions of wafer warpage. By doing so, it enables the development of a simulation model for predicting the wafer warpage induced by back grinding, allowing for the investigation of the effects of process parameters and process time.

In the backside grinding process, the surface of the wafer is damaged by the grinding wheel, which generates residual stress in the surface. Cross-sectional observations of the microcracks

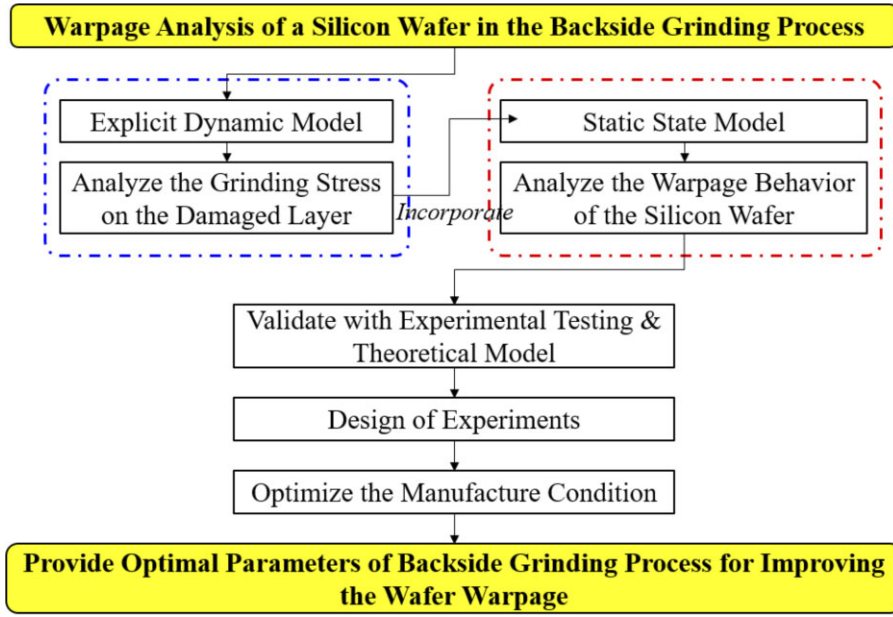


Figure 3 Flowchart of the research process.

on the damaged layer were presented by Gao *et al.* [10]. Then, the grinding stress obtained from the explicit dynamic model is applied to the static state model. Therefore, the predictive model of wafer warpage for the grinding process is developed by using the explicit dynamic model and the static state model. A design of experiments (DOE) approach is used to validate the simulation modeling

4. THEORETICAL MODEL

The grinding force exerted on the silicon wafer by the grinding wheel can be separated into the normal force and the horizontal force, which can be expressed as follows [8]:

$$F_{tt} = \left\{ N\mu \left\{ \frac{5.08R_e^2}{\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}} \right\} + 8.08N\phi k \cdot R_e^2 \right\} \cdot \frac{(r \cdot f \cdot N_w)^{0.6}}{(L \cdot W \cdot \gamma)^{0.6} \cdot N_s^{1.2} \cdot \left(1 + \frac{r^2}{8R_s^2}\right)^{0.6}}, \quad (1)$$

$$F_{nt} = N \left\{ \left\{ \frac{5.08R_e^2}{\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}} \right\} + 8.08k \cdot R_e^2 \right\} \cdot \frac{(r \cdot f \cdot N_w)^{0.6}}{(L \cdot W \cdot \gamma)^{0.6} \cdot N_s^{1.2} \cdot \left(1 + \frac{r^2}{8R_s^2}\right)^{0.6}}, \quad (2)$$

where F_{tt} is the total tangential grinding force, F_{nt} is the total normal grinding force, L is the circumference of the diamond segment, W is the width of the diamond segment, R_e is the equivalent radius of the abrasive grain, E_1 is the Young's modulus of the wafer, E_2 is the Young's modulus of the abrasive grain, ν_1 is Poisson's ratio of the wafer, ν_2 is Poisson's ratio of the abrasive grain, ϕ is the ratio of tangential force to normal force, μ is the

friction coefficient, k is the chip thickness coefficient, r is the radius of the wafer, N_s is the rotational speed of the wheel, N_w is the rotational speed of the wafer, f is the feed rate, R_s is the radius of the wheel and N is the number of effective abrasive grains.

The grinding stress is generated when the grinding force is applied to the wafer. It can be observed that the equations earlier-mentioned include a multitude of process and structural parameters. Therefore, the theoretical model can fully describe the grinding force on the silicon wafer and help validate the simulation model. The theory expresses the relevant influence parameters of the grinding force in detail, so it can be used as a reference for simulating stress generation.

5. FINITE ELEMENT MODELING

The simulation approach is summarized here in two steps. Step 1 determines the stress on the wafer due to the grinding wheel, and step 2 is to apply stress from step 1 to determine the wafer warpage. The explicit dynamic model is composed of a grinding wheel and a silicon wafer, as shown in Fig. 4. And, for the material properties, please check the following Table 1. SOLID 164 elements (8-node brick elements) are used in the dynamic and explicit analysis. SOLID164 is used for the three-dimensional solid structures with eight nodes that have translation, velocity, and acceleration freedom. The number of elements in the grinding wheel is 400, and the number of nodes is 2200. Since the hardness of the grinding wheel is much greater than that of the silicon wafer, the grinding wheel is defined as a rigid body. The grinding wheel moves downward while rotating. Additionally, the silicon wafer is meshed with SOLID 164 elements, and convergence analysis of the established model is carried out to determine the optimal mesh size. The finite element model of the silicon wafer contains 6912 elements and 14 018 nodes. Because the silicon wafer is fixed on a vacuum chuck during the grinding process, a constrained boundary condition is applied to the bottom of the

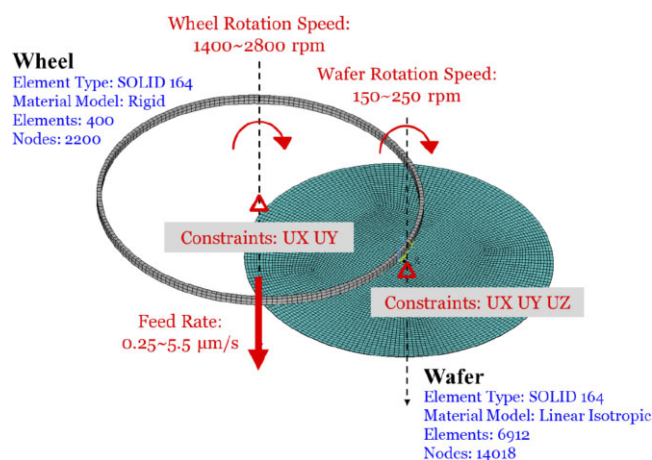


Figure 4 The structure of the explicit dynamic model.

Table 1 Material properties of diamond wheel and silicon wafer

	Density (kg/mm^3)	Young's modulus (GPa)	Poisson ratio	Elements
Diamond wheel	3.51×10^{-6}	1130	0.2	400
Silicon wafer	2.33×10^{-6}	131	0.28	6912

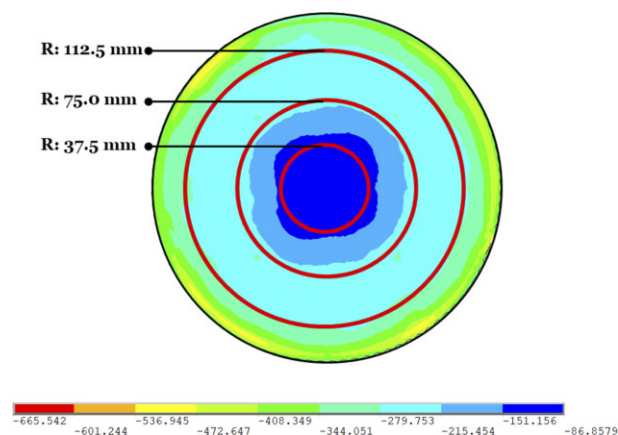


Figure 5 The stress distribution in the silicon wafer.

silicon wafer. In the explicit dynamic model, the feed rate, grinding wheel rotational speed, wafer rotational speed and grinding time are investigated to analyse the grinding stress on the silicon wafer. To enhance the efficiency and accuracy of the simulation model, the grinding process is usually divided into a rough grinding process and a fine grinding process, and the corresponding process parameters are different. An explicit dynamic model is established to calculate the two-stage process and predict the total stress of the wafer through the principle of superposition. The calculation time required for the simulation model established in our study is approximately half an hour. As shown in Fig. 5, the grinding stress on the silicon wafer is generated from the center to the edge. The simulation results show that the minimum grinding stress arises at the center of the wafer and that the maximum grinding stress occurs at the edge of the wafer. For a silicon wafer with a diameter of 300 mm, the grinding stress is measured at radii of 37.5, 75 and 112.5 mm.

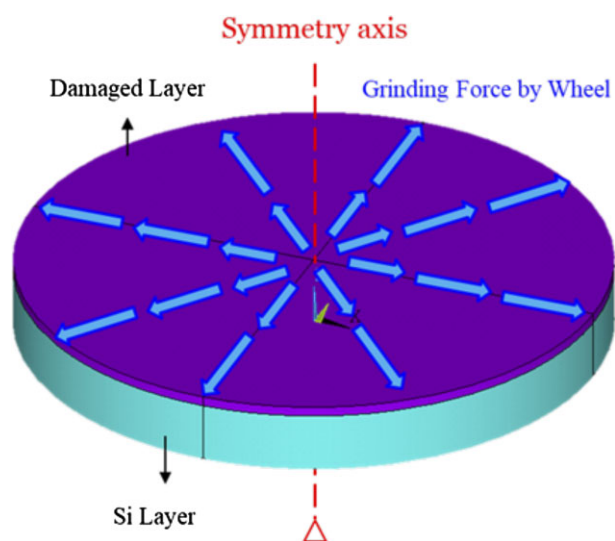


Figure 6 The structure of the static state model.

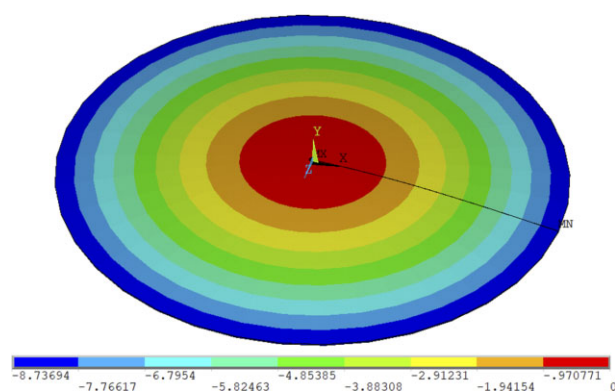


Figure 7 The warpage of the silicon wafer.

To investigate the warpage of silicon wafers in the grinding process, a static state model is developed with two-dimensional axisymmetric, as shown in Fig. 6. The damaged layer of the wafer contains scratches and micro cracks. During the grinding process, the wafer and the grinding wheel are in close contact, so the stress generated by the damage will occur in the damaged layer. These forms of damage are the main reason for the residual stress in the wafer surface. The strength and material properties of the damaged layer are different from those of the no damaged layer due to the occurrence of cracks. The grinding force occurs during the grinding process and is distributed on the damaged surface of the wafer. The silicon wafer is warped by the residual stress in the damaged layer, as shown in Fig. 7.

In the explicit dynamic simulation, the stress on the surface of the wafer is captured and imported into the steady state model of the wafer to calculate the wafer warpage. In the simulations, the grinding process has a feed rate of $1.9 \mu\text{m/s}$, a wheel rotational speed of 1750 rpm and a wafer rotational speed of 250 rpm. The warpage contour is circular and convex due to the uniformity of the grinding force.

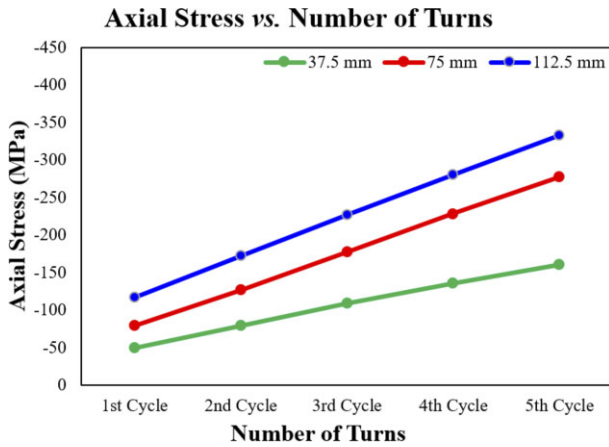


Figure 8 The residual stress in the silicon wafer with respect to the number of grinding cycles in the explicit dynamic model.

6. RESULTS

In the present study, the explicit dynamic model is used to analyse the history of residual stress in the silicon wafer during the backside grinding process. As shown in Fig. 8, the stress is obtained from three index positions at radii of 37.5, 75 and 112.5 mm. The stress in the wafer surface increases as the number of turns increases, and the trend is very stable. The residual stress in the wafer after the grinding process can be predicted from the slope of the trend. In this research, one cycle is defined as one revolution of the wafer. In this case, the thickness of the wafer is ground from 780 to 360 μm , and the overall grinding process will go through approximately 500 cycles. To increase the efficiency of the simulation calculation, the slope of the axial stress with respect to the number of grinding cycles is calculated to obtain the axial stress of the final cycle. The slope is calculated with the following equation:

$$m = \frac{B_2 - B_1}{A_2 - A_1}, \quad (3)$$

where m is the slope of the axial stress rising rate, B_2 is the grinding stress at the final cycle, B_1 is the grinding stress at the first cycle, A_2 is the final number of cycles, and A_1 corresponds to the first cycle.

To achieve the high efficiency silicon wafer warpage prediction, we verify by the relationship between the silicon wafer residual stress growth trend and grinding time. According to Fig. 9, it points out when the grinding time increases, the residual stress shows a linear trend with the increase of the grinding process. By the confirmation of this tendency, this study can predict the residual stress on the surface of the wafer by a more efficient way of a shorter time simulation.

A. One-factor design

To compare the influence of each parameter and wafer warpage, three process parameters—wheel rotational speed, feed rate and wafer rotational speed—are tested with three levels of parameter values using the single-factor analysis method. In addition, the value of each process parameter from the first to the fifth turns of the wheel grinding is captured, and the growth and distribu-

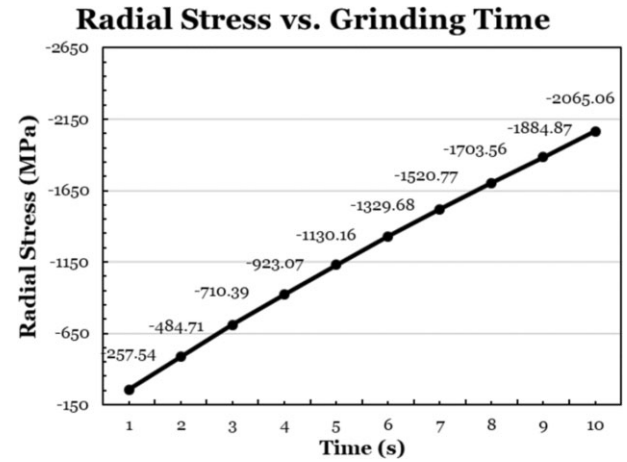


Figure 9 The relationship between radial stress and grinding time.

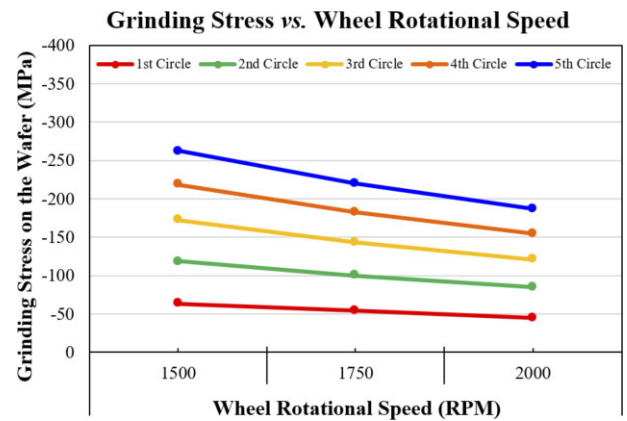


Figure 10 The grinding stress at a radius of 112.5 mm for each cycle with different wheel rotational speeds in the simulation.

tion of grinding stress on the surface of the wafer are observed. In the simulation model, the axial stress on the wafer is defined as the grinding stress. To clearly present the trend of the grinding stress, the grinding stress in the radius of 112.5 mm is analysed. The grinding stresses at a radius of 112.5 mm with different wheel rotational speeds are shown in Fig. 10. As the wheel rotational speed increases, the grinding stress on the wafer surface will decrease. The main factor is that as the grinding wheel speed increases, the grinding depth and the number of effective grinding abrasive particles will decrease, which means the better surface roughness. Consequently, the grinding force will significantly decrease. The grinding force is directly proportional to the wafer speed and feed rate because the grinding depth and damage rate will gradually increase with the progress of the process time and the grinding wheel feed rate. When the wafer speed increases, the grinding speed of the wafer surface and the grinding wheel in contact will also increase, resulting in an increase in grinding force. The cycles of grinding stress at the radius of 112.5 mm with different wafer rotational speeds are shown in Fig. 11 for each cycle. As the wafer rotational speed increases, the grinding stress also increases. When the wafer rotational speed increases, the force on the surface of the wafer is uneven, which causes an increase in grinding stress on the wafer. The cycles of grinding stress at a radius of 112.5 mm with different feed rates

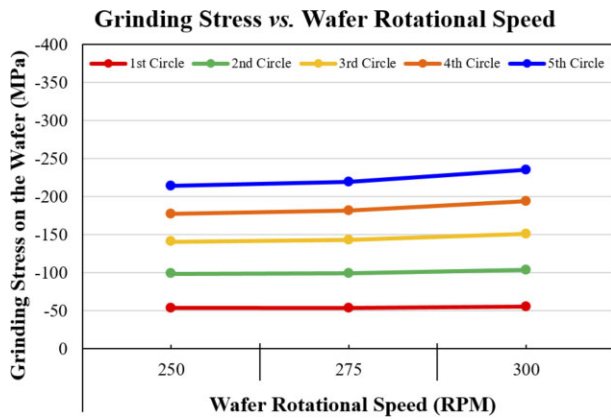


Figure 11 The grinding stress at a radius of 112.5 mm for each cycle with different wafer rotational speeds in the simulation.

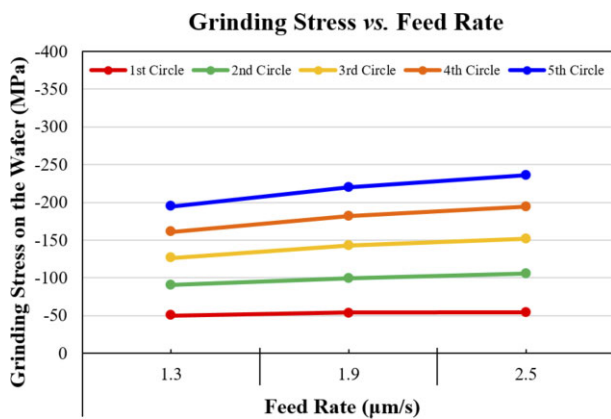


Figure 12 The grinding stress at a radius of 112.5 mm for each cycle with different feed rates in the simulation.

are shown in Fig. 12. When the feed rate increases, the grinding stress will also increase. According to the results from the Finite Element Methodology (FEM) model, the wheel rotation speed is the most important factor affecting the grinding stress. The wafer rotation speed and the feed rate are the least important factors. By the theory of the grinding force, as the wheel rotational speed increases, the grinding depth and the number of effective grinding abrasive particles will decrease, which means the better surface roughness. Consequently, the grinding force will significantly decrease.

B. Theoretical model vs. explicit dynamic model

To compare the relationship of grinding tangent force and grinding stress, the results of the theoretical model and simulation model in the same situation are discussed. As shown in Fig. 13, when the wheel rotational speed increases, the grinding tangent force applied on the wafer will decrease. Because as the wheel rotational speed increases, the grinding depth and the number of effective grinding abrasive grains cutting the wafer decrease, thereby decreasing the grinding tangent force. The grinding tangent force is proportional to the wafer rotational speed and the feed rate because the grinding depth and destruction rate increase as the feed rate increases.

According to Stoney's equation [12], the residual stress on the silicon wafer is investigated in the radial direction. Therefore, the

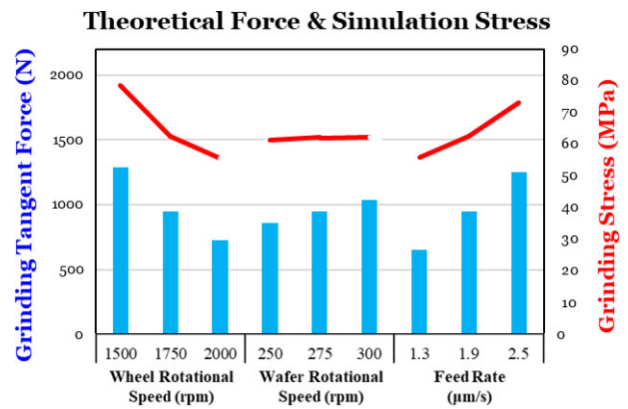


Figure 13 The grinding tangent force (experimental testing) and the grinding stress (simulation) with varying wheel rotational speeds, wafer rotational speeds, and feed rates.

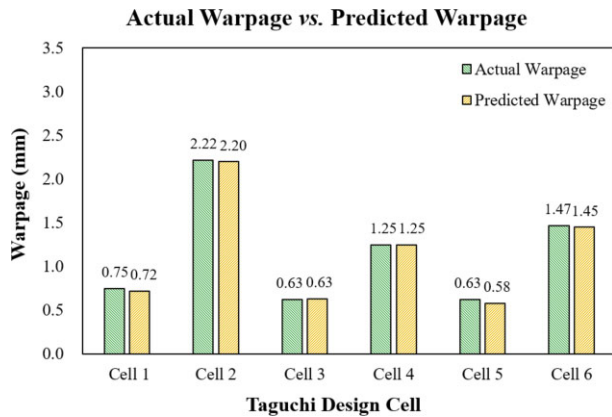
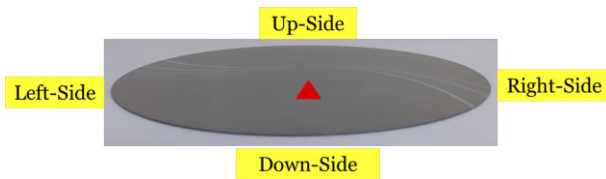
axial stress is determined as the analytical index in the explicit dynamic model. Under the same parameter conditions, the wafer grinding stress in the dynamic simulation model has a trend similar to that of the grinding tangent force in the theoretical model. Because stress is defined as force per unit area, it can be used to investigate the influence between the grinding tangent force and grinding stress. The physical properties of the wafer in the grinding process can be verified through simulation and theoretical verification. Since the stress on the wafer surface is difficult to measure, theoretical calculations can help understand the trend of wafer warpage.

C. Design of experiment

The backside grinding process is divided into the rough grinding stage (Z_1) and the fine grinding stage (Z_2), as shown in Table 2. The grinding process causes cracks on the surface of the wafer. Therefore, the Z_1 stage is carried out to quickly grind the wafer to a thickness close to the target thickness. Next, the Z_2 stage to reduces the roughness of the wafer surface caused by grinding with a lower feed rate. In this study, the Taguchi design is applied to ensure good performance in the design stage. To explore the influence of the process parameters, including the wheel rotational speed, wafer rotational speed, and feed rate, each parameter is tested at two levels of values. In the Taguchi design, the two levels for the three process parameters in the Z_1 and Z_2 stages are randomly combined into six cells. The experimental design for the selection of process parameters in this study follows the L6 experimental format. There are a total of six experimental groups in this experimental table, allowing for the inclusion of five sets of parameter designs with two levels each as the following of Table 2. The actual warpage in the experimental tests and the warpage predicted in the simulations are compared for each cell. The predicted and actual warpage results of the Taguchi design is shown in Fig. 14. A multifactor DOE is conducted to validate the model established by the study, and analyse the effects of various process parameters on warpage based on the results. A photograph of the wafer after grinding in cell 1 is shown in Fig. 15, and the surface shows an arc-shaped scratched surface. Wafer warpage is measured by using the wafer's center as the reference point and measuring

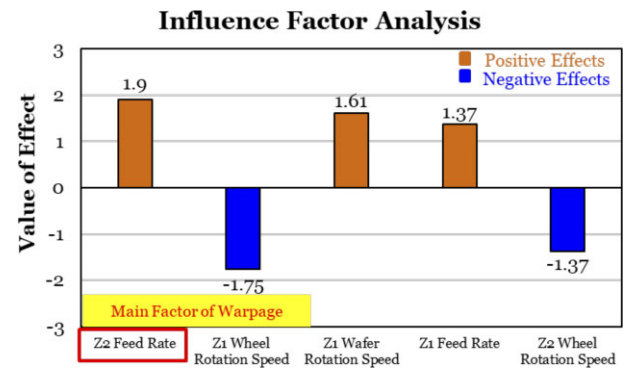
Table 2 Taguchi method of the backside grinding process

Cell	Rough grinding (Z_1 process)			Fine grinding (Z_2 process)			Results		
	Wheel rotational speed (rpm)	Wafer rotational speed (rpm)	Feed rate ($\mu\text{m/s}$)	Wheel rotational speed (rpm)	Wafer rotational speed (rpm)	Feed rate ($\mu\text{m/s}$)	Actual warpage (mm)	Predicted warpage (mm)	Error (%)
1	2000	−150	5.5	2200	−250	0.25	0.75	0.72	4.00
2	2000	−250	1.5	2200	−250	0.35	2.22	2.20	0.85
3	2800	−250	1.5	2200	−250	0.35	0.63	0.63	0.80
4	2800	−250	5.5	1400	−250	0.25	1.25	1.25	0
5	2000	−150	1.5	1400	−250	0.25	0.63	0.58	0.58
6	2800	−150	5.5	1800	−250	0.35	1.47	1.45	1.28

**Figure 14** Actual wafer warpage vs. predicted warpage.**Figure 15** Specimen of silicon wafer after the grinding process.

the height between the warped wafer and the reference plane at four positions: top, bottom, left, and right. These measurements serve as the basis for determining the wafer's warpage. After averaging the measurements taken at the four positions, the average wafer warpage value is obtained. The value of wafer stress in the model linearly increases with time. The average error between the predicted and actual warpage is 3%. According to the following Fig. 3, the results of the analysis using the Taguchi experimental design method provide the factor influence rates of the process parameters on wafer warpage. From the Fig. 16, it can be observed that the grinding wheel feed rate (Z_1) in the fine grinding process has the most significant impact on wafer warpage. Therefore, when optimizing the wafer warpage, adjusting the factors that have a significant influence can help improve the degree of wafer warpage.

After Taguchi design analysis based on Design Expert analysis software, the optimization parametric design is calculated, and we optimize the parameters according to the prediction model established in this study. Increasing the wheel rotational speed and decreasing the wafer rotational speed and feed rate can effectively decrease the grinding stress between the grinding wheel

**Figure 16** Schematic of wafer warpage measurement.**Table 3** Optimal parameters of the backside grinding process

Condition	Standard	Optimization
Z_1 wheel rotational speed (rpm)	2400	2800
Z_1 feed rate ($\mu\text{m/s}$)	3.5	1.5
Z_1 wafer rotational speed (rpm)	150	100
Z_2 wheel rotational speed (rpm)	1800	2200
Z_2 feed rate ($\mu\text{m/s}$)	0.3	0.25
Z_2 Wafer rotational speed (rpm)	300	250
Warpage (mm)	0.69	0.43

and the wafer. The optimized process parameters of the backside grinding process are shown in Table 3.

7. CONCLUSION

This paper analyses the warpage and grinding stress of a 300 mm silicon wafer with a thickness of $360\ \mu\text{m}$ during the backside grinding process. Most studies have used Stoney's equation to calculate the residual stress in silicon wafers via warpage measurements. Other studies have derived a theoretical model for calculating the grinding force by analysing the grinding mechanism. In this study, the accuracy of the simulation model is verified through a Taguchi experimental design. Based on the parameter combinations of the grinding process, the optimal process parameter design is proposed. Therefore, a static model incorporated with an explicit dynamic model is developed herein to simulate the grinding stress on the damaged layer of the silicon wafer and predict the induced warpage. The results from the simulation model are in good agreement with the

experimental results. Increasing the wheel rotational speed, decreasing the feed rate, and decreasing the wafer rotational speed can reduce the wafer warpage. Finally, the optimal parameters of the manufacturing process are presented, which enable a 37.7% reduction in wafer warpage compared with the standard parameters.

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